Data

1. ☐ The type of thé graphic is adapted to the nature of data (curve, bars, pie, histogram, cloud...) ;
2. ☐ Approximations/interpolation make sense ;
3. ☐ Curves are defined by a sufficient number of points ;
4. ☐ The building method of the curve is clear : interpolation (linear, polynomial,regression...) ;
5. ☐ Confidence intervals are visualized (or given separately) ;
6. ☐ Steps of histograms are adequate ;
7. ☐ Histograms visualize probabilities (from 0 to 1).

Graphical objects

1. ☐ Graphical objects are readable on screen, on printed version (B/W), on video... ;
2. ☐ Graphic range is standart, without too similar colors, without green (video) ;
3. ☐ Graphical axis are well identified and labelled ;
4. ☐ Scales and units are explicits ;
5. ☐ Curves cross without ambiguity ;
6. ☐ Grids help the reader.

Annotations

1. ☐ Axis are labelled by quantities ;
2. ☐ Labels of the axis are clear, and self contained ;
3. ☐ Units are indicated on the axis ;
4. ☐ Axes are oriented from the left to the right and from the bottom to the top ;
5. ☐ Origin is $(0, 0)$, if not it should be clearly justified ;
6. ☐ No hole on the axes.

Annotations (2)

6. ☐ For bar graphs/histograms order of bars is based on classical ordering (alphabetical, temporal, from the best to the worse) are better than a random order ;
7. ☐ Each curve has a legend ;
8. ☐ Each bar has a legend ;

Information

1. ☐ Curves are on the same scale ;
2. ☐ The number of curves on a same graph is small (less than 6) ;
3. ☐ Compare curves on a same graphic ;
4. ☐ A curve cannot be removed without reducing the information ;
5. ☐ The graphic gives a relevant information to the reader ;
6. ☐ If the vertical axis shows averages, it should indicates error bars ;
7. ☐ It is not possible to remove any objet without modifying the readability of the graphic.

Context

1. ☐ All the symbols are defined and referenced in the text ;
2. ☐ The graphic produces more information than any other representation (choice of the variable) ;
3. ☐ The graphic has a title ;
4. ☐ The title is sufficiently self contained to partially understand the graphic ;
5. ☐ The graphic is referenced in the text ;
6. ☐ The text comment the figure.

Last but not least : **The graphical representation should be elegant**